1. A local library needs a website to streamline operations and enhance user experience. The site will include a catalog for searching and viewing books, user accounts for borrowing and managing books, and an admin dashboard for managing inventory and user data. Essential features include a search function, book details page, user login system, and contact form, The system helps both students and library managers to keep constant track of all the books available in the library. It allows both the admin and the student to search for the desired book. It becomes necessary for colleges to keep a continuous check on the books issued and returned and even calculate fines.

Librarian Admin:

Login

Add Book

Update/Delete Book

Issue Book for student

Return Book Status

Make Fine

Update material (year, semester wise)

View material

User

Login

Register

Search Book

My Borrow History Book

o Issue Date, Exp Date

Check Fine

Search material (year, semester wise)

Download material

My profile

Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Library Management System</title>
<style>
body {
   font-family: Arial;
   background: #eef2f7;
   margin: 0;
}
header {
   background: #263b80;
   color: white;
   padding: 20px;
   text-align: center;
```

```
}
nav {
  background: #172554;
  padding: 12px;
  text-align: center;
}
nav button {
  padding: 10px 18px;
  margin: 5px;
  border: none;
  cursor: pointer;
  background: white;
  border-radius: 5px;
}
.container {
  width: 85%;
  margin: 25px auto;
  background: white;
  padding: 20px;
  border-radius: 10px;
}
input, select {
  padding: 10px;
  margin: 7px;
  width: 90%;
}
button.action {
  background: #263b80;
  color: white;
  padding: 10px 15px;
  border: none;
  cursor: pointer;
  border-radius: 5px;
}
.book {
  background: #f0f4ff;
  padding: 12px;
  margin: 10px;
  border-radius: 6px;
}
section { display: none; }
#home { display: block; }
</style>
</head>
```

```
<body>

<header>
<h1>Library Management System</h1>
<p>Online Book Search and Management</p>
</header>

<nav>
<button onclick="show('home')">Home</button>
<button onclick="show('books')">Search Books</button>
<button onclick="show('login')">Login</button>
<button onclick="show('borrow')">Borrow History</button>
<button onclick="show('materials')">Materials</button>
<button onclick="show('admin')">Admin</button>
</nav>

<div class="container">

<section id="home">
<h2>Welcome to Our Library</h2>
<p>Search books, borrow books and access study materials online.</p>
<h3>Library Services</h3>
<ul>
<li>Online Book Search</li>
<li>Book Issue and Return</li>
<li>Fine Calculation</li>
<li>Study Material Download</li>
<li>Student Account Management</li>
</ul>
</section>

<section id="books">
<h2>Search Books</h2>
<input type="text" id="search" placeholder="Enter book name"
onkeyup="searchBook()">
<div id="bookList"></div>
</section>

<section id="login">
<h2>Student Login / Registration</h2>
<input type="text" id="username" placeholder="Student Name">
<input type="email" placeholder="Email Address">
<input type="password" placeholder="Password">
<button class="action" onclick="login()">Login / Register</button>
```

```
<p id="loginMsg"></p>
</section>

<section id="borrow">
<h2>My Borrow History</h2>
<table border="1" cellpadding="10">
<tr>
<th>Book Name</th>
<th>Issue Date</th>
<th>Due Date</th>
<th>Fine</th>
</tr>
<tr>
<td>Python Programming</td>
<td>01-11-2026</td>
<td>10-11-2026</td>
<td>₹0</td>
</tr>
</table>
<h3>Fine Calculator</h3>
<input type="number" id="days" placeholder="Enter late days">
<button class="action" onclick="fine()">Calculate Fine</button>
<p id="fineResult"></p>
</section>

<section id="materials">
<h2>Study Materials</h2>
<select id="year">
<option>First Year</option>
<option>Second Year</option>
<option>Third Year</option>
<option>Fourth Year</option>
</select>
<select id="semester">
<option>Semester 1</option>
<option>Semester 2</option>
<option>Semester 3</option>
<option>Semester 4</option>
<option>Semester 5</option>
<option>Semester 6</option>
<option>Semester 7</option>
<option>Semester 8</option>
</select>
<button class="action" onclick="material()">Search Material</button>
```

```
<p id="materialResult"></p>
</section>

<section id="admin">
<h2>Admin Dashboard</h2>
<input type="text" id="newBook" placeholder="Enter Book Name">
<input type="text" id="author" placeholder="Enter Author Name">
<button class="action" onclick="addBook()">Add Book</button>
<h3>Manage Books</h3>
<div id="adminList"></div>
</section>

</div>

<script>
let books = [
{ name:"Python Programming", author:"John Smith" },
{ name:"Java Programming", author:"James Gosling" },
{ name:"Data Structures", author:"Mark Allen" },
{ name:"Web Technology", author:"David" },
{ name:"Database Management", author:"Korth" }
];

function show(id) {
   document.querySelectorAll("section").forEach(s => {
      s.style.display = "none";
   });
   document.getElementById(id).style.display = "block";
   if(id === "books") searchBook();
   if(id === "admin") adminBooks();
}

function searchBook() {
   let value = document.getElementById("search").value.toLowerCase();
   let result = books.filter(b =>
      b.name.toLowerCase().includes(value) ||
      b.author.toLowerCase().includes(value)
   );

   document.getElementById("bookList").innerHTML =
      result.map(b =>
      `<div class="book">
      <h3>${b.name}</h3>
      <p>Author: ${b.author}</p>
```

```
        <button class="action" onclick="issueBook('${b.name}')">
        Issue Book</button>
        </div>`
        ).join("") || "No books found.";
}

function issueBook(name) {
    let student = document.getElementById("username").value;
    if(!student) {
        alert("Please enter your name in Login / Registration.");
        show("login");
        return;
    }
    alert(name + " issued successfully to " + student);
}

function login() {
    let name = document.getElementById("username").value;
    if(name.trim() === "") {
        alert("Please enter your name");
        return;
    }
    document.getElementById("loginMsg").innerHTML =
        "Welcome " + name + "! Login successful.";
}

function fine() {
    let days = Number(document.getElementById("days").value);
    let amount = days > 0 ? days * 5 : 0;
    document.getElementById("fineResult").innerHTML =
        "Total Fine Amount: ₹" + amount;
}

function material() {
    let y = document.getElementById("year").value;
    let s = document.getElementById("semester").value;
    document.getElementById("materialResult").innerHTML =
        "Materials available for " + y + " - " + s;
}

function addBook() {
    let name = document.getElementById("newBook").value;
    let author = document.getElementById("author").value;
```

```
    if(name.trim() === "" || author.trim() === "") {
        alert("Enter book name and author");
        return;
    }

    books.push({name:name, author:author});
    document.getElementById("newBook").value = "";
    document.getElementById("author").value = "";
    adminBooks();
    alert("Book added successfully!");
}

function adminBooks() {
    document.getElementById("adminList").innerHTML =
        books.map((b,i) =>
        `<div class="book">
        ${b.name} - ${b.author}
        <button class="action" onclick="deleteBook(${i})">
        Delete</button>
        </div>`
        ).join("");
}

function deleteBook(i) {
    books.splice(i,1);
    adminBooks();
}
</script>

</body>
</html>
```

Output:

